THOMAS V. THOMPSON II, PH.D.

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PROFESSIONAL SUMMARY

Senior Software Engineer with extensive expertise architecting high-performance C++ systems, complex data pipelines, and scalable procedural geometry engines. Creator of industry-standard XGen and 3-time Academy Scientific and Technical Award nominee. Proven track record of reducing complex technical challenges into robust systems by distilling problems down to their simplifying fundamentals and bridging offline and real-time workflows.

SELECTED CAREER HIGHLIGHTS

- **Industry-Standard Software Creation:** Authored XGen, the procedural instancing framework for hair, fur, and foliage now integrated into Autodesk Maya and used across films, VFX, and game development.
- **Academy SciTech Recognition:** 3-time Academy Scientific and Technical Award nominee for the architectural innovation and industry impact of XGen, iGroom, and Bonsai.
- **Patented Geometry Innovation:** Awarded U.S. Patent for key procedural geometry connectivity and interpolation (“Determining Guide Connectivity”).
- **Pipeline Architecture:** Designed Belaro, Blizzard’s end-to-end, groom-agnostic hair rendering and animation pipeline for cinematics.

TECHNICAL SKILLS

Languages & Tools: C++, Python, C, Qt, PySide, PyQt, Git, Perforce, Linux

Engine & Pipeline Architecture: Unreal Engine 5, Katana, Maya API, XGen, Alembic, USD, Hair/Fur Pipelines

Rendering & Graphics: RenderMan, Redshift, Procedural Geometry, Instancing, OpenGL

Core Competencies: Systems Architecture, Distributed Computation, Technical Leadership, Mentoring

PROFESSIONAL EXPERIENCE

Senior Tools Engineer III

Lightspeed LA — Irvine, CA [2023 - 2026]

- Led technical integration of narrative cinematics in Unreal Engine 5, developing custom tools for importing, updating, and synchronizing Sequencer content.
- Engineered hair and groom conversion tools to translate XGen grooms into tileable flow maps for efficient real-time fur shaders.
- Streamlined studio-wide rendering workflows for frame-accurate cinematic exports and implemented virtual camera (Vcam) capture and export tooling.

Senior Core Software Engineer

Imagendary Studios — Irvine, CA [2022 - 2023]

- Served as technical lead for the Core Software team, designing cinematic workflows, DCC integrations, asset data pipelines, and remote production systems.
- Engineered tools to export XGen grooms and shot caches from Maya into Unreal Engine 5.

Senior Software Engineer II

Blizzard Entertainment — Irvine, CA [2012 - 2022]

- Architected Belaro, an end-to-end groom-agnostic hair rendering and animation pipeline that replaces using live XGen when rendering. This cuts down on data transfer, networking, and computation at rendertime.
- Designed and engineered MobBoss (mass instanced animated crowd pipeline) and Scatter (cross-department geometry instancing solution).
- Authored render-time auto-instancing for Katana to drastically reduce scene memory footprint; served as studio SME for procedural geometry and liaison to Autodesk, Pixar, and Redshift.

Software Supervisor / Senior Software Engineer

Walt Disney Animation Studios — Burbank, CA [2001 - 2012]

- Lead architect of XGen, the studio's arbitrary instancing engine used to manage massive datasets for hair, fur, foliage, and environment layout.
- Patented key procedural connectivity algorithms; authored interactive grooming toolset (iGroom), procedural foliage generator (Bonsai), and unifying image manipulation libraries (ImageToolkit).
- Supervised Lookdev/FX tools engineering team of 20+ developers; delivered specialized technical workshops onsite at Disney, Pixar, ImageMovers Digital, and Digital Domain.

Earlier: Research Staff, University of Utah — real-time geometry, haptic rendering, and distributed computation.

EDUCATION

Ph.D., Computer Science
University of Utah
Salt Lake City, UT
Dissertation: "Haptic Rendering of Trimmed NURBS Models within an Active Prototyping Environment"

M.S., Computer Science
Wake Forest University
Winston-Salem, NC
Thesis: "Three Dimensional Polygon-Mesh Morphology"

B.S., Computer Science
West Virginia Institute of Technology
Montgomery, WV
Honors: Presidential Leadership Award, Alpha Chi